Stratified care in hip arthroscopy: can we predict successful and unsuccessful outcomes? Development and external temporal validation of multivariable prediction models

Lasse Ishøi, K. Thorborg, T. Kallemose, J. Kemp, M. Reiman, M. Nielsen, P. Hölmich

Sports Orthopaedic Research Center-Copenhagen (SORC-C), Arthroscopic Center, Department of Orthopedic Surgery, Hvidovre Hospital, Copenhagen University Hospital, Hvidovre, Denmark lasse.ishoei@regionh.dk.

Journal:	Publication Year:	Citations:	DOI:
British Journal of Sports Medicine	2023.0	7	10.1136/bjsports-2022-105534

CLINICAL & AGENTIC AI TAKEAWAY

Common clinical variables can predict the probability of having an unsuccessful outcome 1 year after hip arthroscopy, aiding shared decision-making between surgeon and patient.

ABSTRACT

Objective: Although hip arthroscopy is a widely adopted treatment option for hip-related pain, it is unknown whether preoperative clinical information can be used to assist surgical decision-making to avoid offering surgery to patients with limited potential for a successful outcome. We aimed to develop and validate clinical prediction models to identify patients more likely to have an unsuccessful or successful outcome 1 year post hip arthroscopy based on the patient acceptable symptom state. Methods: Patient records were extracted from the Danish Hip Arthroscopy Registry (DHAR). A priori, 26 common clinical variables from DHAR were selected as prognostic factors, including demographics, radiographic parameters of hip morphology and self-reported measures. We used 1082 hip arthroscopy patients (surgery performed 25 April 2012 to 4 October 2017) to develop the clinical prediction models based on logistic regression analyses. The development models were internally validated using bootstrapping and shrinkage before temporal external validation was performed using 464 hip arthroscopy patients (surgery performed 5 October 2017 to 13 May 2019). Results: The prediction model for unsuccessful outcomes showed best and acceptable predictive performance on the external validation dataset for all multiple imputations (Nagelkerke R2 range: 0.25-0.26) and calibration (intercept range: -0.10 to -0.11; slope range: 1.06-1.09), and acceptable discrimination (area under the curve range: 0.76-0.77). The prediction model for successful outcomes did not calibrate well, while also showing poor discrimination. Conclusion: Common clinical variables including demographics, radiographic parameters of hip morphology and self-reported measures were able to predict the probability of having an unsuccessful outcome 1 year after hip arthroscopy, while the model for successful outcome showed unacceptable accuracy. The externally validated prediction model can be used to support clinical evaluation and shared decision making by informing the orthopaedic surgeon and patient about the risk of an unsuccessful outcome, and thus when surgery may not be appropriate. © Author(s) (or their employer(s)) 2023. No commercial re-use. See rights and permissions. Published by BMJ.

Keywords: Hip, Groin, Arthroscopy, Sports Medicine, Surveys And Questionnaires

Study Overview & Background

Objective: Although hip arthroscopy is a widely adopted treatment option for hip-related pain, it is unknown whether preoperative clinical information can be used to assist surgical decision-making to avoid offering surgery to patients with limited potential for a successful outcome. We aimed to develop and validate clinical prediction models to identify patients more likely to have an unsuccessful or successful outcome 1 year post hip arthroscopy based on the patient acceptable symptom state. Methods: Patient records were extracted from the Danish Hip Arthroscopy Registry (DHAR). A priori, 26 common clinical variables from DHAR were selected as prognostic factors, including demographics, radiographic parameters of hip morphology and self-reported measures. We used 1082 hip arthroscopy patients (surgery performed 25 April 2012 to 4 October 2017) to develop the clinical prediction models based on logistic regression analyses. The development models were internally validated using bootstrapping and shrinkage before temporal external validation was performed using 464 hip arthroscopy patients (surgery performed 5 October 2017 to 13 May 2019). Results: The prediction model for unsuccessful outcomes showed best and acceptable predictive performance on the external validation dataset for all multiple imputations (Nagelkerke R2 range: 0.25-0.26) and calibration (intercept range: -0.10 to -0.11; slope range: 1.06-1.09), and acceptable discrimination (area under the curve range: 0.76-0.77). The prediction model for successful outcomes did not calibrate well, while also showing poor discrimination. Conclusion: Common clinical variables including demographics, radiographic parameters of hip morphology and self-reported measures were able to predict the probability of having an unsuccessful outcome 1 year after hip arthroscopy, while the model for successful outcome showed unacceptable accuracy. The externally validated prediction model can be used to support clinical evaluation and shared decision making by informing the orthopaedic surgeon and patient about the risk of an unsuccessful outcome, and thus when surgery may not be appropriate. © Author(s) (or their employer(s)) 2023. No commercial re-use. See rights and permissions. Published by BMJ.

Methodology & Clinical Focus

Study Classification: other. Evaluated under orthopedic machine learning and artificial intelligence decision frameworks.

Key Findings & Analysis

Common clinical variables can predict the probability of having an unsuccessful outcome 1 year after hip arthroscopy, aiding shared decision-making between surgeon and patient.

Extended Repository Abstract

Objective Although hip arthroscopy is a widely adopted treatment option for hip-related pain, it is unknown whether preoperative clinical information can be used to assist surgical decision-making to avoid offering surgery to patients with limited potential for a successful outcome. We aimed to develop and validate clinical prediction models to identify patients more likely to have an unsuccessful or successful outcome 1 year post hip arthroscopy based on the patient acceptable symptom state. Methods Patient records were extracted from the Danish Hip Arthroscopy Registry (DHAR). A priori, 26 common clinical variables from DHAR were selected as prognostic factors, including demographics, radiographic parameters of hip morphology and self-reported measures. We used 1082 hip arthroscopy patients (surgery performed 25 April 2012 to 4 October 2017) to develop the clinical prediction models based on logistic regression analyses. The development models were internally validated using bootstrapping and shrinkage before temporal external validation was performed using 464 hip arthroscopy patients (surgery performed 5 October 2017 to 13 May 2019). Results The prediction model for unsuccessful outcomes showed best and acceptable predictive performance on the external validation dataset for all multiple imputations (Nagelkerke R 2 range: 0.25-0.26) and calibration (intercept range: -0.10 to -0.11; slope range: 1.06-1.09), and acceptable discrimination (area under the curve range: 0.76-0.77). The prediction model for successful

outcomes did not calibrate well, while also showing poor discrimination. Conclusion Common clinical variables including demographics, radiographic parameters of hip morphology and self-reported measures were able to predict the probability of having an unsuccessful outcome 1 year after hip arthroscopy, while the model for successful outcome showed unacceptable accuracy. The externally validated prediction model can be used to support clinical evaluation and shared decision making by informing the orthopaedic surgeon and patient about the risk of an unsuccessful outcome, and thus when surgery may not be appropriate.

Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

Bibliographic Details & Indexing

- **Authors:** Lasse Ishøi, K. Thorborg, T. Kallemose, J. Kemp, M. Reiman, M. Nielsen, P. Hölmich
- **Journal / Venue:** British Journal of Sports Medicine (2023.0)
- **SJR Metric:** Quartile 1.0
- **Citation Count:** 7 citations recorded
- **Digital Object Identifier (DOI):** https://doi.org/10.1136/bjsports-2022-105534
- **Consensus Evidence Link:** https://consensus.app/papers/stratified-care-in-hip-arthroscopy-can-we-predict-ish%C3%B8i-thorborg/44795465ac7c580ea13d07967dd2231c/

Source & Provenance: Curated for Agentic Orthopedics Research Collection • DOI: https://doi.org/10.1136/bjsports-2022-105534 • Consensus Page: https://consensus.app/papers/stratified-care-in-hip-arthroscopy-can-we-predict-ish%C3%B8i-thorborg/44795465ac7c580ea13d07967dd2231c/ • Verified and rendered via Deep Search & Academic Repositories.